

University Course Handbook

Generated System

March 6, 2026



Kalasalingam Academy of Research and Education

(Deemed to be University)

Under Sec. 3 of UGC Act 1956

Anand Nagar, Krishnankoil (Via),
Virudhunagar (Dt) – 626126, Tamil Nadu

Programme Core Mandatory



Kalasalingam Academy of Research and Education

(Deemed to be University)

Under Sec. 3 of UGC Act 1956

Anand Nagar, Krishnankoil (Via),
Virudhunagar (Dt) – 626126, Tamil Nadu

ECE000 ENGINEERING ELECTROMAGNETICS

ECE000 ENGINEERING ELECTROMAGNETICS	L	T	P	X	C
	3	0	2	0	4.0
Pre-requisite: Basic Physics at I year	Category / Type: PCM / IC-T				

Course Description

The course is a fundamental course where students will be learning about the concepts of vector calculus, electrostatics and magnetostatics, the fundamental laws that govern Maxwell's equations. Electromagnetic explains universal concepts in three-dimension real world, i.e., electromagnetic wave propagation in free-space. The course also introduces various effects of EMI/EMC with standard for describing their properties. The course ends with a few insights on applications of EM waves.

Course Objectives

- To expose the students to the rudiments of electromagnetic theory and wave propagation essential for subsequent courses on transmission lines, waveguides, microwave engineering, antennas, and wireless communication.

Course Outcomes

- CO1: Apply vector calculus concepts along with physical principles and properties to effectively solve problems encountered in everyday life, further study in science, and in the professional world related to electric field.
- CO2: Visualise the variations in Magnetic field by the knowledge gained from Ampere's circuital law.
- CO3: Interpret phasor Maxwell's equations in differential and integral forms, in time domain from the electrodynamics concept.
- CO4: Analyse the boundary conditions between the two different mediums in the magnetic field and apply the concepts of uniform plane wave propagation for the time varying electric and magnetic fields.
- CO5: Make appropriate trade-offs to achieve the most cost-effective sustainable design that meets all requirements with the knowledge gained on EMI/EMC due to EM waves.
- CO6: As a team acquire data from computer simulations to explore electromagnetic principles, effectively communicate the results, and critically evaluate the related engineering concepts, following the safety procedures and ethics in the laboratory.

CO-Blooms Mapping

CO	Course Outcome	Bloom's Level
CO1	Apply vector calculus concepts along with physical principles and properties to effectively solve problems encountered in everyday life, further study in science, and in the professional world related to electric field.	K2 - AP
CO2	Visualise the variations in Magnetic field by the knowledge gained from Ampere's circuital law.	K2 - AP
CO3	Interpret phasor Maxwell's equations in differential and integral forms, in time domain from the electrodynamics concept.	K2 - AP
CO4	Analyse the boundary conditions between the two different mediums in the magnetic field and apply the concepts of uniform plane wave propagation for the time varying electric and magnetic fields.	K2 - AP
CO5	Make appropriate trade-offs to achieve the most cost-effective sustainable design that meets all requirements with the knowledge gained on EMI/EMC due to EM waves.	K2 - AP
CO6	As a team acquire data from computer simulations to explore electromagnetic principles, effectively communicate the results, and critically evaluate the related engineering concepts, following the safety procedures and ethics in the laboratory.	K2 - AP



Kalasalingam Academy of Research and Education

(Deemed to be University)

Under Sec. 3 of UGC Act 1956

Anand Nagar, Krishnankoil (Via),
Virudhunagar (Dt) – 626126, Tamil Nadu

Syllabus

legacydocxsource: true

Unit 1: Electrostatics

Vector analysis, Cartesian, Cylindrical and Spherical coordinate systems, Relationship between the coordinate systems, Properties of coordinate system - Coulomb's law and its applications, Electric field intensity, Electric flux densities for various charge distributions such as line, surface, and volume - Gauss's law and its applications - Electrostatic potential, Divergence theorem - Convection and Conduction current - Electric field in free space, conductors, and dielectrics, Electrostatic boundary value problems: Poisson's and Laplace's equations and solutions

Unit 2: Magnetostatics

Magnetic field intensity - Bio Savart's law - Ampere's circuit law - Magnetic flux and Magnetic flux density, Magnetic flux density in free space, conductor, and magnetic materials - Magnetic force, Magnetic Vector potential - Stroke's theorem.

Unit 3: Electromagnetics

Faraday's Law - Maxwell's equations in point form and integral form for steady fields - Boundary conditions using Maxwell's equations - Retarded potentials.

Unit 4: Uniform Plane Wave

Uniform plane wave - Wave polarization - Wave propagation in free space, dielectrics and conducting medium - Power flow and Poynting vector - Plane Waves at a Media Interface,

Unit 5: Applications

EMI / EMC, EMI Sources - Effects of EMI – Testing Methods for EMI/EMC, Methods to suppress EMI-Grounding and shielding, EMC standards - Practical issues of EMI/EMC non compliances, Biological effects of EMI / EMC – ESD - EMP

Laboratory Experiments

1. Represent vectors in various Coordinate system using MATLAB; and Sketch the Surface in Cartesian Coordinates, Cylindrical Coordinates and Spherical Coordinates using MATLAB.
2. Translate vectors between coordinate systems using MATLAB.
3. Find electric flux of charged surfaces and electric field due to point charges, line, surface, and volume charge density using MATLAB.
4. Find Divergence and Curl for a Current Carrying Wire using MATLAB
5. Find the magnetic field from the current carrying conductors using MATLAB
6. Find and plot the normalized vectors E and D in anisotropic Dielectric medium using MATLAB.
7. Visualise Maxwell's equations using MATLAB.
8. Display the charge distribution of Parallel plate capacitor computed by Method of moments using MATLAB
9. Find the Poynting vector of a standing wave and compute the power of plane waves from the pointing vector using MATLAB.
10. Compute dielectric-dielectric boundary conditions for arbitrarily positioned (Oblique) boundary between medium 1 and 2 using MATLAB
11. Analyse EMI/EMC of cables using MATLAB.
12. Design, simulate and analyse a basic monostatic pulsar RADAR system using MATLAB.

Textbook(s)

1. William H. Hayt, John A. Buck, Jaleel M. Akhtar, "Engineering Electromagnetics", 9th Edition, McGraw Hill India, 2020.
2. Mathew O Sadiku, "Elements of Electromagnetics", 6th Edition, Oxford University Press, 2015.

Reference(s)

1. Griffiths, David J, "Introduction to Electrodynamics", 4th Edition, Pearson India, 2015.
2. Dikshitulu K. Kalluri, "Electromagnetic Waves, Materials, and Computation with MATLAB", 2nd Edition, CRC Press, 2018.
3. Edward C. Jordan, Keith G. Balmain, "Electromagnetic Waves and Radiating Systems", 2nd Edition, Pearson India, 2015.



Kalasalingam Academy of Research and Education

(Deemed to be University)

Under Sec. 3 of UGC Act 1956

Anand Nagar, Krishnankoil (Via),
Virudhunagar (Dt) – 626126, Tamil Nadu

-
4. Constantine A. Balanis, “*Advanced Engineering Electromagnetics*”, 2nd Edition, Wiley US, 2015.
 5. Saroj K. Dash, Smruti R. Khuntia., “*Fundamentals of Electromagnetic Theory*”, 2nd Edition, PHI, 2015.
 6. R K Shevgaonkar, “*Electromagnetic Waves*”, 1st Edition, McGraw Hill India, 2015.
 7. <https://www.edx.org/xseries/mitx-introductory-electricity-and-magnetism>
 8. <https://www.coursera.org/specializations/electrodynamics>
 9. https://www.tutorialspoint.com/electromagnetics_theory/index.asp

Administrative Details

Course Revision	1.0
Course Author	Department Curriculum Committee
BoS Approval	Jan/2025
Document Version	N/A
Document Date	N/A
Document Git Hash	N/A



Kalasalingam Academy of Research and Education

(Deemed to be University)

Under Sec. 3 of UGC Act 1956

Anand Nagar, Krishnankoil (Via),
Virudhunagar (Dt) – 626126, Tamil Nadu

ECE001 DIGITAL SIGNAL AND IMAGE PROCESSING

ECE001 Digital Signal and Image Processing	L	T	P	X	C
	3	0	2	0	4.0
Pre-requisite: Signals and Systems	Category / Type: PCM / IC-T				

Course Description

This course aims to provide students with a detail knowledge on design of linear and nonlinear filters for 1D and 2D signals; Understand Finite word length effect; Understand the concepts of various image processing concepts in real time applications

Course Objectives

- To study about the design of IIR and FIR filters
- To study about standard image processing algorithms; Understanding of image processing system development
- To study the extraction and detection of images

Course Outcomes

- CO1: To represent various dimensional signals mathematically in continuous and discrete form in time and frequency domain
- CO2: To design the FIR digital filters for various applications meeting the requirements
- CO3: To design the IIR digital filters for various applications meeting the requirements
- CO4: To design frequency domain filters and spatial filters for image enhancement
- CO5: To analyse the methodologies for image restoration, image segmentation and image compression
- CO6: Work as part of a team and as individual effectively in designing the communication systems following the norms and ethics in practice

CO-Blooms Mapping

CO	Course Outcome	Bloom's Level
CO1	To represent various dimensional signals mathematically in continuous and discrete form in time and frequency domain	K2 - AP
CO2	To design the FIR digital filters for various applications meeting the requirements	K2 - AP
CO3	To design the IIR digital filters for various applications meeting the requirements	K2 - AP
CO4	To design frequency domain filters and spatial filters for image enhancement	K2 - AP
CO5	To analyse the methodologies for image restoration, image segmentation and image compression	K2 - AP
CO6	Work as part of a team and as individual effectively in designing the communication systems following the norms and ethics in practice	K2 - AP

Syllabus

legacydocxsource: true

Unit 1: 1D, 2D Transforms

Signals (1D, 2D, 3D, Multidimensions), 1-D DFT, General Orthonormal Transform, 2-D DFT, 1-D DCT, 2-D DCT, Discrete Sine Transform, Hadamard Transform, KL Transform, Wavelet Transform

Unit 2: FIR Filters

Concept of FIR/IIR – Design Linear Phase FIR filter – Fourier Series Method, LPF Design of FIR Digital filters: Window method - Rectangular, Hamming, Hanning and Blackman window, Frequency sampling method.

Unit 3: IIR Filters

Digital design using impulse invariant and bilinear transformation Warping, pre-warping. Discrete time IIR filter from analog filter-Butterworth and Chebyshev Filter, Filter design (LPF, HPF, BPF, BRF) using frequency translation.



Kalasalingam Academy of Research and Education

(Deemed to be University)

Under Sec. 3 of UGC Act 1956

Anand Nagar, Krishnankoil (Via),
Virudhunagar (Dt) – 626126, Tamil Nadu

Unit 4: Image Enhancement and Image Interpolation

Image Enhancement: Grayscale Transformation, Contrast Stretching, Gamma Correction, Histogram Equalization, Unsharp Masking, Image Interpolation: Linear Interpolation, Polynomial interpolation. Edge Detection: Definition of an Edge-Finite Differences, Roberts Edge Detector, Sobel and Prewitt Edge Detectors, Marr-Hildreth Edge Detection, Canny Edge Detector.

Unit 5: Pixel Coding And DSP Processor

Huffman Coding, LZW Coding, DPCM of Image - JPEG: Standard, Case Study on Vehicle license plate character recognition and Water marking. Introduction to DS Processors – Architecture – Features – Addressing Formats – Functional modes.

Laboratory Experiments

1. Representation of Magnitude and Phase spectrum using 2D-DFT by using MATLAB / equivalent software.
2. Representation of Magnitude and Phase spectrum using 2D-DCT and DST by using MATLAB / equivalent software.
3. Analysis of 2D signals like an Image using DWT by using MATLAB / equivalent software.
4. Design and implement of Basic filters using MATLAB / equivalent software
5. Design and implement of FIR filter using Rectangular and Hamming window for the given specifications using MATLAB / equivalent software
6. Design and implement of FIR filter using Frequency sampling method for the given specifications using MATLAB / equivalent software
7. Design and implement of IIR filter using bilinear transformation, and impulse invariant transformation by using MATLAB / equivalent software.
8. Design and implementation of filter using Butterworth filter and Chebyshev filter using MATLAB / equivalent software.
9. Design of various filters with frequency transformation technique using MATLAB
10. Poor contrast image enhancement using Histogram Equalization in MATLAB / equivalent software.
11. Edge Detection using Sobel and Prewitt operators for the given image using MATLAB / equivalent software.
12. Detection of edges using Canny edge detection method by using MATLAB / equivalent software.
13. Watermarking creation and extraction using MATLAB / equivalent software.
14. Calculating FFT and convolution using DSP processor 'TMS320C6748

Textbook(s)

1. Digital Signal Processing, "Principles, Algorithms and Application," 1st Edition, John G Proakis, 2020.
2. Digital Image Processing, "Rafael C., Gonzalez and Richard E. Woods", 4th Edition, Pearson, 2017.
3. Digital Signal Processors, Architecture, Implementations and Applications, "Sen-Maw Kuo, Woon-Seng Gan", 1st Edition, Pearson, 2020.

Reference(s)

1. Theory and Applications of Digital Signal Processing, "Lawrence R. Rabiner and Bernard Gold", 1st Edition, Pearson, 2020.
2. Digital Image Processing, "William K. Pratt", 4th Edition, John Wiley, 2016.
3. Massimo Tistarelli, Christophe Champod, "Handbook of Biometrics for Forensic Science", 1st Edition, Springer International Publishing, 2017.
4. Digital Signal Processing– A Computer based, "approach", 1st Edition, Sanjit Mitra, 2020.
5. Digital Signal Processing using Matlab, "Vinay K.Ingale, John.G.Proakis," 1st Edition, Asoke K. Gosh, 2015.
6. Digital signal processing using matlab for Students and researchers, "John w. Leis, University of Southern Queensland," 1st Edition, A John Wiley and sons, 2015.
7. http://121.200.55.237/pluginfile.php/8763/mod_resource/content/1/1-Basics%20and%20FFT.pdf
8. http://121.200.55.237/pluginfile.php/8765/mod_resource/content/1/2-IIR.pdf
9. http://121.200.55.237/pluginfile.php/8773/mod_resource/content/1/3-FIR.pdf
10. <https://web.stanford.edu/class/ee368/>



Kalasalingam Academy of Research and Education

(Deemed to be University)

Under Sec. 3 of UGC Act 1956

Anand Nagar, Krishnankoil (Via),
Virudhunagar (Dt) – 626126, Tamil Nadu

Administrative Details

Course Revision	1.0
Course Author	Department Curriculum Committee
BoS Approval	Jan/2025
Document Version	N/A
Document Date	N/A
Document Git Hash	N/A